

Review- 1

Students should download research papers relevant to their chosen concept/title from Scopus and Web of Science (WoS) indexed databases. Only downloadable, full-text journal papers published from 2021 and directly related to your proposed concept should be selected.

Ensure that a valid, working downloadable link is available for each paper, and that the dataset, methodology, and limitations discussed in each paper are clearly identifiable, as these will be tabulated below.

S.No	Name of the paper and Authors	Year of Publication	Downloadable Link	Journal Name / Scopus or Web of Science	Dataset Used	Methodology Used	Limitations
1	Blind Spot Detection Radar System Design for Safe Driving of Smart Vehicles — Wantae Kim, Heejin Yang, Jinhong Kim	2023	https://doi.org/10.3390/app13106147	Applied Sciences (MDPI) — Scopus & Web of Science (SCIE)	No public benchmark dataset reported. Simulation data and custom real-world vehicle tests using the developed BSD radar system.	FMCW radar; ADC; 1D/2D FFT; range-Doppler mapping; peak detection; angle estimation; target sorting and tracking; α - β / α - β - γ tracking filters; custom antenna and modem design; vehicle-level testing.	Complex driving environments make multi-target prioritization difficult. Equal tracking time for all targets can cause tracking failures. The paper identifies AI-based target prioritization as future work.
2	Blind-Spot Monitoring System using Lidar — Kuldeep S. Pawar, Shivanand N. Teli, Prasad Shetye, Saukshit Shetty, Vedant Satam, Atul Sahani	2022	https://doi.org/10.1007/s40032-022-00856-2	Journal of The Institution of Engineers (India): Series C — Scopus	No public benchmark dataset reported. Results were obtained from on-road tests in regular traffic using the developed LiDAR prototype.	Retrofittable wireless BSD using TF-Luna LiDAR, Arduino UNO, HC-05 Bluetooth and BLiM mobile application. Distance-based detection with LED and buzzer alerts. Real-time on-road testing.	Additional sensors are needed to cover different blind-spot zones. The approach is mainly distance-based and does not provide the richer range/velocity/angle information available from FMCW radar. Testing was limited to the developed prototype and reported scenarios.
3	A Multi-Scale Mask Convolution-Based Blind-Spot Network for Hyperspectral Anomaly Detection — Zhiwei Yang, Rui Zhao, Xiangchao Meng, Gang Yang, Weiwei Sun, Shenfu Zhang, Jinghui Li	2024	https://doi.org/10.3390/rs16163036	Remote Sensing (MDPI) — Scopus & Web of Science (SCIE)	Four real hyperspectral datasets: Pavia, Gainesville, San Diego-1 and Gulfport. Sensors include ROSIS-03 and AVIRIS.	Multi-scale blind-spot convolution; dynamic fusion; spatial-spectral joint module; background feature attention; pixel-shuffle down-sampling; spatial-spectral screening; self-supervised hyperspectral anomaly detection; comparison with nine state-of-the-art methods.	This paper addresses hyperspectral anomaly detection rather than automotive blind-spot detection. Its datasets are hyperspectral images, not automotive radar/LiDAR data, so direct transfer to vehicle BSD is not established.
4	Rearview Camera-Based Blind-Spot Detection and Lane Change Assistance System for Autonomous Vehicles — Yunhee Lee, Manbok Park	2025	https://doi.org/10.3390/app15010419	Applied Sciences (MDPI) — Scopus & Web of Science (SCIE)	Custom (not public) rearview-camera dataset: 12,537 images (7,519 train / 2,511 val / 2,507 test) from urban roads and highways with cars, motorcycles, buses and trucks; 185° fisheye camera (1280×720) on a Kia Carnival. Lane detection tested on 4,872 images.	Sobel edge-pair lane detection with 3rd-order polynomial fitting and Kalman lane tracking; fisheye distortion correction; vanishing-point-based camera calibration update to compensate vertical vibration; YOLOv9 (transfer learning) with lane-based ROI for small/distant vehicles; Kalman object tracking; relative distance/speed and time-to-collision; 3-level collision warning. Compared with Faster R-CNN. Result: precision ≈90%,	Camera-only: affected by backlight, glare, blurred lane markings and poor lighting; evaluated mainly in daytime (night, rain and snow left as future work). A rear camera cannot see vehicles immediately adjacent to the car. Lane detection fails at crosswalks/crossroads; distant or backlit cars are missed and a toll gate was misdetected. Recall for cars is lower on urban roads (79.5%). Only

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						recall 82.8%, 11.64 fps (85.87 ms/frame), lane detection rate 91.52%.	≈11.6 fps; distance comes from monocular bounding-box geometry. No radar/LiDAR comparison (future work); dataset is private.
5	Influence of Blind Spot Assistance Systems in Heavy Commercial Vehicles on Accident Reconstruction — Thomas König, Daniel Paula, Stefan Quaschner, Hans-Georg Schweiger	2024	https://doi.org/10.3390/s24051517	Sensors (MDPI) — Scopus & Web of Science (SCIE)	No public dataset. Own field tests at the CARISSMA outdoor test area (60 m × 70 m) with pedestrians, cyclists and a EuroNCAP bicyclist dummy; four test setups (static crossing, static parallel, dynamic parallel, dynamic turning/crossing) following UN-ECE R151 and ADAC procedures.	Experimental comparison of a radar-based system (Mercedes Actros) and a camera-based system (AXION ICA Turn-Assist on a MAN TGM). Measured distance at first information/warning signal vs last information point (LIP), effect of lateral distance, speed ratio, visual obstruction and turn indicator; video and 2D data logger recording; PC-Crash simulation to use the acoustic warning as a reaction prompt point (1.4 s reaction time) in avoidability analysis.	Only two commercial systems on heavy trucks, with few repetitions per test (about three runs); dynamic turning tests only on the radar system (12 runs). Warning-only systems: no active braking. No system detected a VRU behind an obstacle. Radar response was irregular at equal speeds; camera system detects only within ≈2.2 m laterally and mostly when the VRU is faster. Warning timing depends strongly on the turn indicator (sometimes too late). Ultrasonic sensors not tested; wet test surface; data only on request; no event data recorder available.
6	Analysis of Advanced Driver-Assistance Systems for Safe and Comfortable Driving of Motor Vehicles — Tomasz Neumann	2024	https://doi.org/10.3390/s24196223	Sensors (MDPI) — Scopus & Web of Science (SCIE)	No sensor dataset. Paper-based questionnaire survey of 80 drivers (40 women, 40 men, aged 20–30) (7 questions), May–September 2023, on ADAS use, safety and comfort.	Literature-based description of radar, camera, LiDAR and ultrasonic sensors; review of ACC, BSD, LDW/LKS, intelligent headlamp control and emergency brake assist; questionnaire analysis (e.g. >50% reported a significant safety gain, 60% improved comfort); discussion of ADAS failure modes and centralized / distributed / hybrid architectures.	No algorithm or experimental performance evaluation; BSD is covered only descriptively. Small, narrow-age, self-reported sample, so results show perception rather than measured safety. Failure examples are generic and not backed by data. Drivers asked for better precision, fewer false alarms and a better user interface, which this paper leaves unaddressed.
7	Blind Spot Detection for Autonomous Driving Using RADAR Technique — A Y Ramteke, P Ramteke, A Dhongade, U Modak, L P Thakre	2024	https://doi.org/10.1088/1742-6596/2763/1/012015	Journal of Physics: Conference Series (IOP) — Scopus (conference proceedings)	No public dataset. Laboratory prototype tests with a moving object; results shown as Arduino Uno measuring IF (Doppler) frequency with the FreqPeriod library and a frequency threshold; speed = (Doppler shift / 2) × (c / Ft); buzzer alert as HMI.	HB-100 10.525 GHz X-band Doppler radar module; 2-stage LM358 pre-amplifier; Arduino Uno measuring IF (Doppler) frequency with the FreqPeriod library and a frequency threshold; speed = (Doppler shift / 2) × (c / Ft); buzzer alert as HMI.	Doppler-only motion sensing; no range or angle, no object classification or tracking, so it cannot tell which lane a target is in. Threshold-based detection can be triggered by any moving object. Tested only at very low speeds with no on-road vehicle trials or reported accuracy/false-alarm metrics. Background discusses 77 GHz FMCW, but the prototype uses a low-cost 10.5 GHz Doppler module. Conference proceedings rather than a journal.
8	Blind-Spot Collision Detection System for Commercial Vehicles Using Multi Deep CNN Architecture — Muhammad Muzammel, Mohd Zuki Yusoff, Mohamad Naufal Mohamad Saad, Faryal Sheikh, Muhammad Ahsan Awais	2022	https://doi.org/10.3390/s22166088	Sensors (MDPI) — Scopus & Web of Science (SCIE)	Self-recorded blind-spot vehicle detection dataset for buses (not public) plus the public online LISA vehicle detection dataset.	Two CNN-based feature extractors integrated with Faster R-CNN for blind-spot vehicle detection; fusion of pre-trained ResNet-50 and ResNet-101 features. Reported false detection rates of 3.05% and 3.49% on the self-recorded dataset.	Vision-only (camera) detection of vehicles for buses and heavy vehicles: no range/velocity measurement, tracking or time-to-collision, so no risk level is computed. Performance depends on lighting and weather. Two-stage Faster R-CNN is heavier than single-stage detectors. Self-recorded

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							dataset is not public. (Assessed from the abstract and online summary; confirm in the full text.)
9	Recent Advances in mmWave-Radar-Based Sensing, Its Applications, and Machine Learning Techniques: A Review — A. Soumya, C. Krishna Mohan, Linga Reddy Cenkeramaddi	2023	https://doi.org/10.3390/s23218901	Sensors (MDPI) — Scopus & Web of Science (SCIE)	Review article, so no dataset of its own. It surveys mmWave radar models, applications (automotive, industrial, medical, robotics, security) and ML methods applied to radar data.	Review of mmWave FMCW radar principles and metrics (range, velocity, angle of arrival, resolutions, IF signal, range-Doppler and range-azimuth heatmaps, 3D point cloud); radar types (short/medium/long range); automotive applications including blind spot detection, lane change assistance and ADAS; table of machine-learning techniques used with radar data (CNN, LSTM, PointNets, Faster R-CNN, SVM, etc.).	No experiments or new method. Blind spot detection is mentioned only briefly among many applications, with no BSD benchmark or comparison. Interference between radars and weather effects are named as open challenges. Many of the ML examples are non-automotive (gestures, medical, industrial).
10	Forward Collision Warning Strategy Based on Millimeter-Wave Radar and Visual Fusion — Chenxu Sun, Yongtao Li, Hanyan Li, Enyong Xu, Yufang Li, Wei Li	2023	https://doi.org/10.3390/s23239295	Sensors (MDPI) — Scopus & Web of Science (SCIE)	Experiments under various weather and road conditions (details of the data are in the full text; not stated in the abstract).	Decision-level radar-camera fusion: adaptive Kalman filter for radar target tracking with an information-entropy-based memory index; YOLOv5s improved with an SKBAM attention module for vehicle detection; minimum-safe-distance model for the warning. Reduces false alarm rate by 11.619% and missed alarm rate by 15.672% versus the traditional algorithm.	Designed for forward collision warning, not blind-spot detection or side/rear zones. Gains are reported relative to a baseline and rely on a camera detector that degrades in poor visibility. Safe-distance warning has no multi-level risk prioritization of several targets. (Assessed from the abstract and online summary; confirm in the full text.)

Note: All ten papers are from journals/proceedings indexed in Scopus. Applied Sciences, Sensors and Remote Sensing are also indexed in Web of Science SCIE; the Journal of Physics: Conference Series paper (Paper 7) is Scopus-indexed conference proceedings. DOI links above provide the paper access pages. A preprint by Waghmare et al. (Preprints.org, July 2026, not peer-reviewed) was reviewed for background but is not tabulated because it is not a Scopus/WoS indexed journal paper.

Based on the datasets, methodologies, and limitations identified from the reviewed literature above, students should clearly articulate how their proposed system differs from and improves upon the existing approaches. This justification should explicitly address the novelty of the proposed work by describing which limitations of the existing methods/datasets are being overcome, what new dataset(s) or methodology are being introduced or adapted, and why the proposed approach is expected to perform better or offer a meaningful contribution over the reviewed papers. (POINT WISE)

Proposed System Novelty and Improvement (Point Wise)

1. Use FMCW radar as the primary sensing technology to obtain range, relative velocity and angle information for moving vehicles. This goes beyond the distance-only LiDAR prototype (Paper 2) and the Doppler-only HB-100 prototype (Paper 7), which give no range or angle.
2. Add AI-based threat assessment to address the target-prioritization limitation identified in the FMCW radar paper (Paper 1).
3. Adapt a public automotive radar dataset such as RadarScenes and generate left/right blind-spot labels from object position, velocity and tracking information. Papers 1, 2, 5 and 7 rely on custom prototypes or field tests, and Paper 4 uses a private camera dataset, so none can be reproduced.
4. Implement multi-target prioritization using distance, relative velocity, position and estimated collision risk.
5. Include Time-to-Collision (TTC) as an additional feature so approaching vehicles can be assigned a higher risk level, and use a graded (multi-level) warning similar to Paper 4 but driven by radar measurements instead of monocular camera geometry.

6. Extend the system with camera-radar sensor fusion: radar provides range/velocity/angle while the camera provides visual object classification. This reduces the weaknesses of the camera-only systems in Papers 4 and 8 (backlight, poor lighting, no direct range/velocity, about 11.6 fps in Paper 4) and of the camera-based system in Paper 5 (small lateral coverage).
7. Apply decision-level radar-camera fusion with a tracking filter (adaptive Kalman) to the left/right blind-spot zones. Paper 10 shows the benefit of this fusion only for forward collision warning, not for blind-spot detection.
8. Build on the FMCW processing chain surveyed in Paper 9 (range-Doppler, angle of arrival, CFAR) and use machine learning on radar data for classification and prioritization. Paper 9 is only a survey and gives no blind-spot benchmark.
9. Monitor both left and right blind-spot zones, addressing the LiDAR paper's observation that additional sensors may be needed for wider coverage.
10. Use radar velocity and tracking to detect targets moving at about the same speed as the ego vehicle, which Paper 5 found to be irregular for the tested radar system, and avoid depending on turn-indicator status for the alert.
11. Reduce false alarms by CFAR detection and track-based filtering. False alarms and precision were the main improvements drivers asked for in the survey in Paper 6.
12. Provide real-time visual/audio driver warnings based on the calculated risk level.
13. Maintain a clear automotive focus: the third paper's 'blind-spot network' is for hyperspectral anomaly detection and is not an automotive BSD system.
14. Expected contribution: combine FMCW radar sensing, target tracking, blind-spot zoning and AI-based threat prioritization to improve reliability in multi-target driving situations.